

Event Causal Analysis (ECA) - AFW Pump P-101A Vibration Alarm

ECA ID: ECA-2026-0007 | Linked CR: CR-2026-00123 | Linked WO: WO-2026-04567 | Status: Final

1. Event Summary

On 2026-02-06, AFW Pump P-101A experienced a vibration alarm during a routine surveillance start. The pump was stopped and declared inoperable pending evaluation. Subsequent maintenance actions restored vibration to acceptable levels.

2. Timeline

Time	Event
09:10	P-101A started for surveillance; flow stabilized at ~465 gpm.
09:11	VB-101A peaked at 6.2 mils; alarm received.
09:12	Operator stopped pump; CR initiated.
02/07	WO executed; alignment checked and corrected.
02/07	Verification run passed; vibration 3.8 mils max.

3. Evidence

Evidence considered includes vibration peak values, flow and pressure readings, suction pressure near lower typical bound, and maintenance findings indicating misalignment corrected during WO execution.

Evidence Item	Observation
VB-101A	Peak 6.2 mils pre-maintenance; 3.8 mils post-maintenance.
Alignment	Offset reduced from 4.1 mils (as-found) to 2.5 mils (as-left).
Process conditions	Flow and discharge pressure nominal; suction pressure lower than typical but stable.

4. Analysis

The vibration alarm is most consistent with mechanical misalignment. Correction of alignment reduced vibration below the alarm threshold without changes to process conditions. Cavitation is considered less likely because flow and pressure remained stable and no noise/instability was documented, although suction margin should be monitored.

5. Causal Factors

Type	Factor
Direct Cause	Coupling misalignment contributed to elevated vibration during start.
Contributing Factor	Recent maintenance activity may have altered alignment tolerance.
Contributing Factor	Suction pressure near lower typical bound reduced margin (monitor).

6. Corrective / Preventive Actions

Corrective: Align coupling and verify vibration below 5.0 mils (completed via WO-2026-04567). Preventive: Add post-maintenance vibration check for P-101A coupling work; review suction pressure trending during surveillance.

Assumptions / Limitations: This example ECA is test data for software pipeline validation. It is not intended for operational decision-making.